

Demonstration of WDM, QoT-Aware Rate Adaptation and Add/Drop Operation in Inter-satellite All-optical Networks

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Abstract Using a seven-node inter-satellite all-optical network (ISAON) testbed driven by Starlink TLE data, we demonstrate WDM, QoT-aware rate adaptation, and WXC add/drop under realistic time-varying inter-satellite loss emulation. Under HPA-limited power, four-channel WDM reaches 800 Gb/s. WDM-capable QoT estimation enables upgrades with ≤ 1.4 dB error. ©2026 The Author(s)

Introduction

Low Earth orbit (LEO) satellite constellations such as Starlink already employ optical inter-satellite links to provide ~ 100 Gbps-class global connectivity [1], but growing traffic and stringent latency requirements expose the limitations of architectures that depend on onboard electrical processing. To address this, an inter-satellite all-optical network (ISAON) has been proposed to eliminate O/E/O regeneration and reduce redundant processing time and power consumption [2–5] (Fig. 1). In ISAON, each satellite is equipped with laser communication terminals (LCTs), a wavelength cross-connect (WXC) switch (SW), and rate-adaptive transceivers (Fig. 1), enabling dynamic wavelength routing akin to terrestrial optical networks.

While ISAON enables WDM path reconfiguration and WXC-based add/drop as in terrestrial

optical networks, three issues remain. (i) With stringent HPA output constraints in satellite payloads, WDM decreases per-channel launch power, and its net capacity gain is not self-evident; our recent numerical study quantified the potential benefit under HPA-limited operation [6]. Here, we experimentally verify this benefit by adaptively selecting the number of wavelengths and data rates under HPA constraints. (ii) Variability of component performance across cascaded nodes (satellites) complicates end-to-end QoT (quality of transmission) prediction and induces excessive margins; we have validated digital-twin-based QoT estimation using optical probing in a single-channel system [4,7]. This paper extends the method to four-channel WDM and demonstrates capacity upgrades with sufficient estimation accuracy. (iii) ISAON also relies on WXC-based add/drop to flexibly allocate relay satellites for multiple source–destination pairs, whose feasibility should be validated under time-varying inter-satellite loss, as demonstrated in this work.

We experimentally demonstrate on a seven-node ISAON testbed using Starlink orbital data: four-channel WDM with jointly adaptive multi-rate assignment under HPA-limited power, WDM-capable digital-twin QoT estimation with capacity upgrades, and WXC-based add/drop operation under time-varying loss.

Experimental setup and WDM-capable QoT algorithm

Fig. 2(a) shows a seven-node ISAON testbed with add/drop capability, based on our proof-of-concept platform in [4] and extended to WDM operation. Between adjacent nodes, VOAs emulate time-varying inter-satellite loss without introducing fiber dispersion or nonlinear effects. The launch power at each node was set to 23 dBm, corresponding to the 33 dBm-class HPA output in [2] after ~ 10 dB coupling loss. An OpenZR+-compliant QSFP-DD DCO transceiver [9] generated 30/60 Gbaud DP-QPSK (100/200 Gb/s) and 60 Gbaud DP-8/16QAM (300/400 Gb/s). Four 100 GHz-spaced WDM channels were emulated by

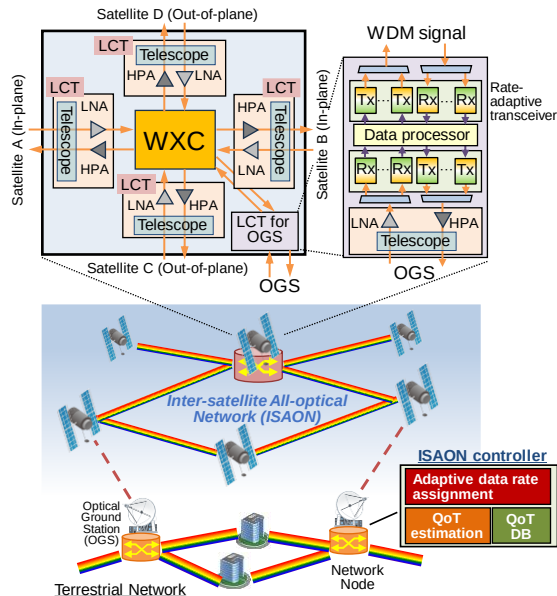


Fig. 1: Concept of inter-satellite all-optical network and configuration of satellite. The rate-adaptive transceiver is connected to WXC, providing add/drop function, which is also linked to the transceiver from the OGS via a data processor. The data processor offers routing capabilities, including data transfer from/to data storage. LCT: Laser communication terminal, LNA: Low-noise amplifier, HPA: High-power amplifier.

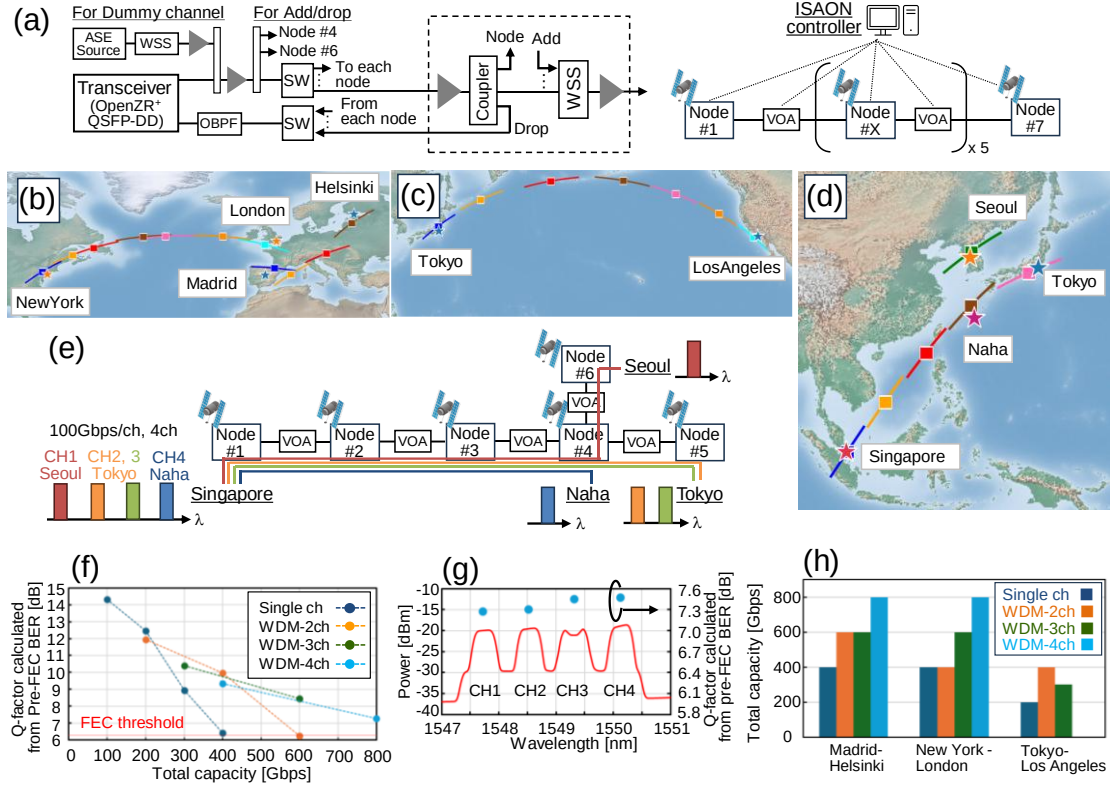


Fig. 2: (a) Experimental setup, (b), (c) ground tracks for adaptive rate assignment and QoT estimation, (d) ground tracks for add/drop experiment, (e) network configuration for add/drop experiment, (f) measured spectrum and Q-factor for the case of New York-London, (g) Q-factor with single and 2-4 channels WDM transmission versus total capacity for the case of New York-London, (h) Total capacity with single and 2-4 channels WDM transmission.

combining one modulated wavelength with ASE-shaped dummy wavelengths, and 1×8 optical switches enabled add/drop at any node. Pre-FEC BER measured by the transceiver was used as the QoT metric.

The controller configured paths by switching the input/output nodes and collecting pre-FEC BER, following [4,7]. Inter-satellite distances were derived from Starlink TLE data ([8], Mar. 6, 2026) and converted to span losses applied to the VOAs using the link-budget formulation in [2], as in [4,7]. We considered three OGS pairs (Madrid–Helsinki, New York–London, and Tokyo–Los Angeles) for adaptive rate assignment and QoT estimation, and a multi-destination add/drop scenario relaying traffic from Singapore via Naha toward Tokyo and Seoul. The ground tracks (projection of the orbit onto Earth’s surface) of the satellites are shown in Figs. 2(b)–(d). Fig. 2(e) depicts the add/drop configuration: a 4×100 Gb/s (400 Gb/s) WDM signal drops CH1 at Naha, CH4 at Seoul, and CH2–CH3 at Tokyo; in the reverse direction, the same wavelengths are added and re-aggregated into 400 Gb/s toward Singapore.

We next outline the WDM-capable QoT estimation for QoT-aware rate adaptation. Transceiver #1 probes each span at 200 Gb/s/λ under two-channel loading (signal + ASE dummy; WDM-2ch) and stores span-level BER in a QoT

database (DB) to build a digital twin. During service, QoT for requested 100–800 Gb/s aggregate WDM paths carried by Transceiver #2 is estimated from the DB (see [7] for details): (i) converts stored probing Pre-FEC BER to per-span generalized SNR, (ii) compensates Transceiver #1 contribution [10] using calibration parameters, (iii) accumulates restored span SNRs to obtain path SNR and rescales it for the target WDM channel count (e.g., +3 dB for single-channel relative to WDM-2ch probing), (iv) incorporates Transceiver #2 contribution at the target rate, and (v) predicts Pre-FEC BER and Q-factor. The controller then assigns the highest data rate meeting $Q \geq 6.3$ dB [9], using the minimum WDM channel count among feasible options, enabling operation closer to the true link capability compared with a conventional design based on equipment specifications with a fixed margin.

Experimental results

We first discuss the adaptive data-rate assignment results using the New York–London case. Fig. 2(f) plots the Q-factor corresponding to the total capacity; for WDM transmission we report the worst-channel Q-factor. At total capacities of 300 and 400 Gb/s, introducing WDM provides a larger performance margin than the single-

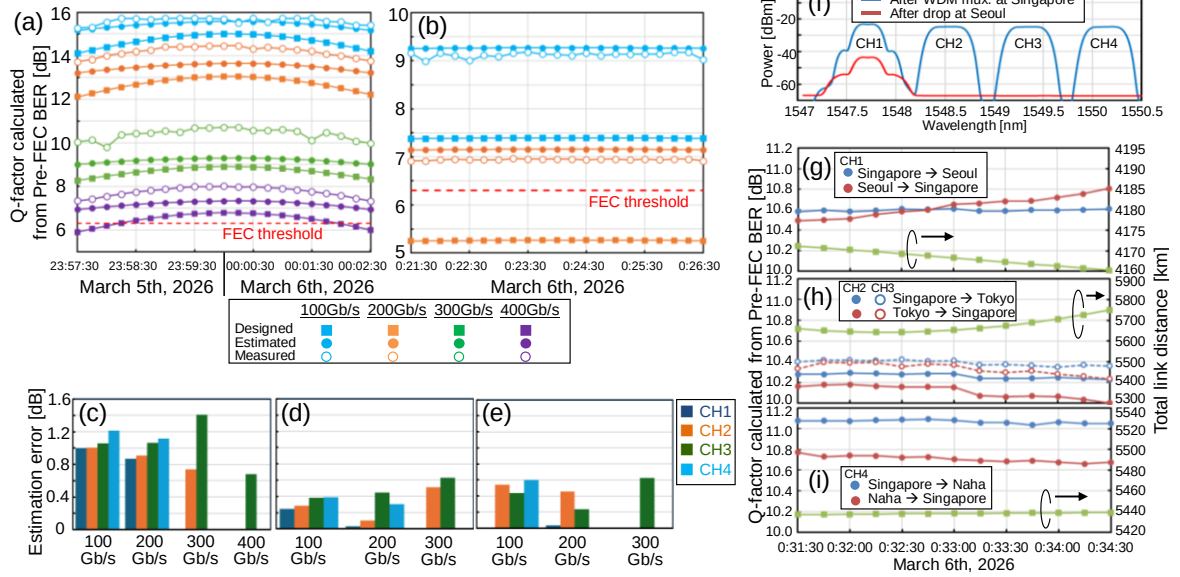


Fig. 3: Estimated, measured, and designed Q-factor for the case of (a) Madrid – Helsinki and (b) Tokyo – Los Angeles. Estimation error for each data rate for the case of (c) Madrid-Helsinki, (d) New York – London and (e) Tokyo – Los Angeles. (f) Spectra at the output of Singapore and after drop at Seoul. (g)-(i) Q-factors of three source-destination pairs for the add/drop experiment.

channel case, indicating that multi-wavelength operation can be sustained under HPA-limited conditions. The maximum total capacity achieved in this experiment was 800 Gb/s, realized with four-channel WDM at 200 Gb/s/λ, and the corresponding measured spectrum and Q-factor are shown in Fig. 2(g). We confirmed that the worst-channel Q-factor remains above the FEC threshold (6.3 dB [9]) even at 800 Gb/s in our conditions. Fig. 2(h) summarizes the assigned capacities for single- and 2–4-channel WDM, confirming that jointly adapting the per-wavelength data rate and the number of WDM channels yields an approximately twofold capacity gain under limited HPA output power.

We next evaluate the WDM-capable QoT-estimation experiment. Figs. 3(a) and 3(b) show the Madrid-Helsinki single-channel case and the Tokyo-Los Angeles two-channel WDM case, respectively. The square symbols indicate the designed Q-factor based on equipment specifications, where the amplifier noise figure was set to 6.5 dB (typ. ~4.5 dB) to include sufficient margin. With this conventional design, the assigned data rate would be 300 Gb/s/λ for Madrid-Helsinki and 100 Gb/s/λ for Tokyo-Los Angeles. In contrast, the proposed digital-twin QoT estimation (filled circles) enables upgrades to 400 Gb/s/λ and 200 Gb/s/λ, respectively; the measured Q-factors (open circles) also exceed the FEC threshold in these upgraded settings. Figs. 3(c)–3(e) summarize the estimation error for each WDM channel (CH1–CH4) across the three path conditions, where the worst-case error is at most 1.4 dB, demonstrating accuracy sufficient to reduce conservative margins and enable capacity upgrades

in WDM operation.

Finally, we discuss the multi-source add/drop demonstration enabled by the WXC. The measured spectra at the output of Singapore and after dropping at Seoul are shown in Fig. 3(f), confirming add/drop operation. Figs. 3(g)–3(i) plot the Q-factors together with the corresponding total link distance versus time for each source-destination pair. The results demonstrate both the drop function that distributes traffic from Singapore to multiple destinations (via Naha toward Tokyo and Seoul) and the add function that aggregates traffic from multiple sources and delivers it to Singapore. These demonstrations indicate the feasibility of efficiently allocating relay satellites for multiple concurrent flows using WXC-based add/drop operation in ISAON under time-varying inter-satellite loss.

Conclusions

We experimentally demonstrated WDM, QoT-aware rate adaptation, and WXC add/drop for inter-satellite all-optical networks using a Starlink-orbit-driven seven-node testbed. Under HPA-limited power, four-channel WDM achieved up to 800 Gb/s. Digital-twin QoT estimation enabled rate/capacity upgrades with ≤1.4 dB worst-case error, supporting efficient multi-traffic-flow operation under time-varying loss.

Acknowledgements

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